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Editors-in-Chief
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Dear Editors,

I am pleased to submit *"The Post-ChatGPT Contraction Reaches the Crowd-Validated Margin: A Within-Tag Triple Difference and a Value-Calibrated Early-Warning System for Stack Overflow"* for consideration at *Decision Support Systems.*

Problem and decision relevance.
Public question-and-answer communities such as Stack Overflow are both a knowledge resource for millions of practitioners and the training substrate for the very generative-AI systems that now compete with them. Recent work establishes that Q&A activity contracted after the November 2022 ChatGPT release (Burtch, Lee, and Chen 2024; del Río-Chanona, Laurentsyeva, and Wachs 2024, ∼15–25% declines) and that the surviving question population is, on average, of higher quality (Xue et al. 2026). A platform manager—or an AI provider monitoring the health of its training corpus—reading those two facts together might conclude that generative AI prunes only low-value posts that public Q&A would not have retained as durable artefacts. If true, no intervention is warranted. The decision-relevant question, which an aggregate quantity-and-quality decomposition cannot answer, is whether the contraction also reaches the *high-value, crowd-validated* posts that sustain the platform's downstream knowledge function—and, if so, how a manager who observes only activity counts should detect and triage the loss. This paper answers both the measurement question and the decision question.

Contributions.

1. *The strict benign-pruning reading is rejected at the within-tag value margin.* A single pre-specified shape test over mutually exclusive net-score bins returns a negative average displacement across the crowd-validated range ($\hat{\beta} = -0.167$, $p < 10^{-4}$) that is roughly constant over the low-to-mid bins (score 0, 1, 2: -0.21, -0.20, -0.21) and attenuates only at the rare elite tail (slope $p = 2 \times 10^{-4}$). The displacement is therefore not confined to score-zero posts.
2. *A within-tag, within-question-type triple difference* at the (tag × week × question-type) cell isolates the LLM-substitutability slice of the aggregate decline, reported under *two co-primary* treatment measures—a historical answerability index (-0.108, $p = 0.021$)

and an embedding-based score that uses no historical features (-0.119, $p < 0.001$)—so the result cannot be dismissed as historical popularity relabelled.

3. *A value-calibrated early-warning system—the decision-support artefact.* We let the data choose the detection signal: out of sample on the full 100-tag panel, the activity-decline metric a platform already tracks predicts realised 2025 high-value erosion at AUC 0.98, while a value-weighted detector reaches only 0.57. The system therefore *detects* on activity, and the paper's value-margin estimates serve as the *calibration* that converts an activity alert into the high-value posts at risk that managers actually care about. The calibration earns its place at the intensive (triage) margin: allocating intervention effort by the calibrated high-value-at-risk nearly attains the perfect-foresight optimum, whereas allocating by raw activity leaves up to 5% of the protectable high-value flow unprotected. A highly persistent composition flag ($\rho = 0.90$ from 2024 into the new 2025 data) classifies whether a tag's decline is value-disproportionate, informing intervention *type*.

Fit with *Decision Support Systems*.
The paper's core deliverable is a decision aid, evaluated as one. The early-warning system is built from signals a platform already collects, is calibrated on identified causal estimates rather than correlations, and is assessed by its out-of-sample detection performance and by a value-of-information / regret analysis that quantifies the cost of the naive activity-only policy a manager would otherwise use. This is squarely the journal's remit: analytics and AI in support of managerial decisions, with an explicit account of the decision the tool improves and the loss it averts. The empirical contribution (the within-tag value-margin estimate) is what makes the calibration trustworthy; the artefact is what makes it actionable.

Identification, stated honestly.
Formal pre-trend tests reject *exact* parallel trends, so we do not claim a standalone natural experiment: the estimates are conditional post-ChatGPT associations aligned with the LLM-substitutability mechanism. Their credibility rests on a conjunction of diagnostics that fail under different threats—a sign reversal between the real cutoff and 29 rolling pre-period placebos, a positive (placebo) June 2022 GitHub Copilot cutoff, monotone growth of the event-study coefficient with no reversion, and a HonestDiD Δ^{RM} sensitivity bound under the full event-study variance–covariance matrix with a breakdown at $\bar{M} = 0.75$, which we read as a *moderate* signal and say so. The estimate also survives a 32-cell pre-specified specification curve (all negative, all significant), PPML, two-way clustering, and a wild cluster bootstrap. Because two one-off 2023 platform events (a data-licensing change and a moderator strike) overlap the early post-period, we treat a donut difference-in-differences that excludes the affected weeks as the *primary* test against that confound (-0.110, $p = 0.022$). Finally, re-estimating on the *full 100-tag panel extended through 2025* strengthens the triple difference to -0.172 ($p = 0.002$) and the quarterly association deepens to -0.68 by the end of 2025 with no reversion, removing the reading that the effect reflects a transient late-2024 episode.

Scope discipline.
We resist overclaiming. The identified channel is bounded at $\sim$4% of post-period activity (and $\sim$5% of a trend-implied shortfall); a five-transition support-post funnel translates it

into a cumulative shortfall of ∼28,000 accepted, non-closed, non-negative-score posts (about 40% of the activity shortfall). The estimate is an innovation *input* rather than an outcome, and the paper states plainly what the magnitude does *not* support (welfare, innovation outcomes, or generalisation beyond programming Q&A). A conditional-resolvability null under fine-grained fixed effects shows that the descriptively documented quality improvement is consistent with selection into the survivor population rather than within-cell upgrading. All references have been verified against Crossref/DOI.

Reproducibility and structure.
The full collection-and-analysis pipeline is reproducible from public, zero-budget data: manual Stack Exchange Data Explorer exports through May 2023 and the public Stack Exchange API v2.3 thereafter (including the 2025 extension), with byte-level validation across the source migration on the overlap period. The manuscript is self-contained within the journal's 34-page limit (double-spaced, 1-inch margins); to keep it within that limit the full robustness battery, the per-bin value-distribution detail, the mechanism and classifier-validation detail, the descriptive counterfactual, and the literature-positioning table are provided in the public replication package rather than as journal-hosted supplementary material. All Python scripts, panels, tables, figures, and the LaTeX source are documented in `docs/replication_notes.md` and available at `https://github.com/hlaverde/so-genai-value-margin`.

Suggested expertise.
Referees with expertise in decision-support and early-warning systems, analytics for platform governance, the economics of generative AI, online knowledge communities, or quasi-experimental methods (triple difference, pre-trend sensitivity bounds, cluster-robust inference, specification curve analysis) would be well positioned to evaluate the work.

Disclosures.
The work is single-authored and has not been submitted to any other journal. There are no conflicts of interest. No paid APIs, restricted institutional data, or external funding were used.

Sincerely,

Henry Laverde-Rojas
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